PREDICTIVE MODELLING PROJECT REPORT

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PROBLEM 1 : LINEAR REGRESSION

You are hired by a company Gem Stones co ltd, which is a cubic zirconia manufacturer. You are provided with the dataset containing the prices and other attributes of almost 27,000 cubic zirconia (which is an inexpensive diamond alternative with many of the same qualities as a diamond). The company is earning different profits on different prize slots. You have to help the company in predicting the price for the stone on the bases of the details given in the dataset so it can distinguish between higher profitable stones and lower profitable stones so as to have better profit share. Also, provide them with the best 5 attributes that are most important.

DATA ATTRIBUTES

Variable Name	Description
Carat :	Carat weight of the cubic zirconia.
Cut :	Describe the cut quality of the cubic zirconia. Quality is increasing order Fair, Good, Very Good, Premium, Ideal.
Color :	Colour of the cubic zirconia.With D being the worst and J the best.
Clarity :	Clarity refers to the absence of the Inclusions and Blemishes. (In order from Worst to Best in terms of avg price) IF, VVS1, VVS2, VS1, VS2, Sl1, Sl2, l1
Depth :	The Height of cubic zirconia, measured from the Culet to the table, divided by its average Girdle Diameter.
Table :	The Width of the cubic zirconia's Table expressed as a Percentage of its Average Diameter.
Price :	the Price of the cubic zirconia.
X :	Length of the cubic zirconia in mm.
Y :	Width of the cubic zirconia in mm.
Z :	Height of the cubic zirconia in mm.

SAMPLE OF THE DATASET

	Unnamed: 0	carat	cut	color	clarity	depth	table	x	y	z	price
0	1	0.30	Ideal	E	SI1	62.1	58.0	4.27	4.29	2.66	499
1	2	0.33	Premium	G	IF	60.8	58.0	4.42	4.46	2.70	984
2	3	0.90	Very Good	E	VVS2	62.2	60.0	6.04	6.12	3.78	6289
3	4	0.42	Ideal	F	VS1	61.6	56.0	4.82	4.80	2.96	1082
4	5	0.31	Ideal	F	VVS1	60.4	59.0	4.35	4.43	2.65	779

Table 1. Original Dataset sample (problem 1)

The dataset has 11 variables but the column 'Unnamed" is not useful for the model so we are dropping it from the dataset and now the dataset has 10 columns.

1.1. Read the data and do exploratory data analysis. Describe the data briefly. (Check the null values, Data types, shape, EDA, duplicate values). Perform Univariate and Bivariate Analysis.

After dropping Unnamed column the new dataset sample having 10 rows is as follows :

	carat	cut	color	clarity	depth	table	x	y	z	price
0	0.30	Ideal	E	SI1	62.1	58.0	4.27	4.29	2.66	499
1	0.33	Premium	G	IF	60.8	58.0	4.42	4.46	2.70	984
2	0.90	Very Good	E	VVS2	62.2	60.0	6.04	6.12	3.78	6289
3	0.42	Ideal	F	VS1	61.6	56.0	4.82	4.80	2.96	1082
4	0.31	Ideal	F	VVS1	60.4	59.0	4.35	4.43	2.65	779

Table 2. New dataset sample (problem 1)

Check for the shape of the dataframe.

The dataset has 26967 rows and 10 columns after dropping Unnamed from the dataset.

Check for the data type.

carat	float64
cut	object
color	object
clarity	object
depth	float64
table	float64
x	float64
y	float64
z	float64
price	int64

Three variables are of object data type while one variable is of integer data type and rest are float.

Check for the missing values in the dataset.

carat	26967 non-null
cut	26967 non-null
color	26967 non-null
clarity	26967 non-null
depth	26270 non-null
table	26967 non-null
x	26967 non-null
y	26967 non-null
z	26967 non-null
price	26967 non-null

Approximately 697 missing values are present in “depth” variable while other attributes don’t have any missing value.

Check for duplicates

There are total 34 duplicate rows and we are dropping these duplicates.
Thus, the resulting shape of the dataset now is 26933 rows and 10 columns.

Summary stats

Descriptive statistics help describe and understand the features of a specific dataset by giving short summaries about the sample and measures of the data. The most recognized types of descriptive statistics are measures of centre : the mean, median and mode.

	carat	cut	color	clarity	depth	table	x	y	z	price
count	26933.000000	26933	26933	26933	26236.000000	26933.000000	26933.000000	26933.000000	26933.000000	26933.000000
unique	NaN	5	7	8	NaN	NaN	NaN	NaN	NaN	NaN
top	NaN	Ideal	G	SI1	NaN	NaN	NaN	NaN	NaN	NaN
freq	NaN	10805	5653	6565	NaN	NaN	NaN	NaN	NaN	NaN
mean	0.798010	NaN	NaN	NaN	61.745285	57.455950	5.729346	5.733102	3.537769	3937.526120
std	0.477237	NaN	NaN	NaN	1.412243	2.232156	1.127367	1.165037	0.719964	4022.551862
min	0.200000	NaN	NaN	NaN	50.800000	49.000000	0.000000	0.000000	0.000000	326.000000
25%	0.400000	NaN	NaN	NaN	61.000000	56.000000	4.710000	4.710000	2.900000	945.000000
50%	0.700000	NaN	NaN	NaN	61.800000	57.000000	5.690000	5.700000	3.520000	2375.000000
75%	1.050000	NaN	NaN	NaN	62.500000	59.000000	6.550000	6.540000	4.040000	5356.000000
max	4.500000	NaN	NaN	NaN	73.600000	79.000000	10.230000	58.900000	31.800000	18818.000000

Table 3. Summary (problem 1)

From the descriptive statistics, we can see that the average price of the cubic zirconia is around 3940 while the maximum price is 18818. The average weight of the diamond is around 0.80 while the maximum weight is 4.5, from the above summary stats we can see that the count in depth attribute is slightly different than other attributes which shows the presence the null values.

Boxplot

Boxplot displays the five-number summary of a dataset. The five number summary is the minimum, first quartile, median, third quartile and maximum. It also displays the outliers proportion in the dataset.

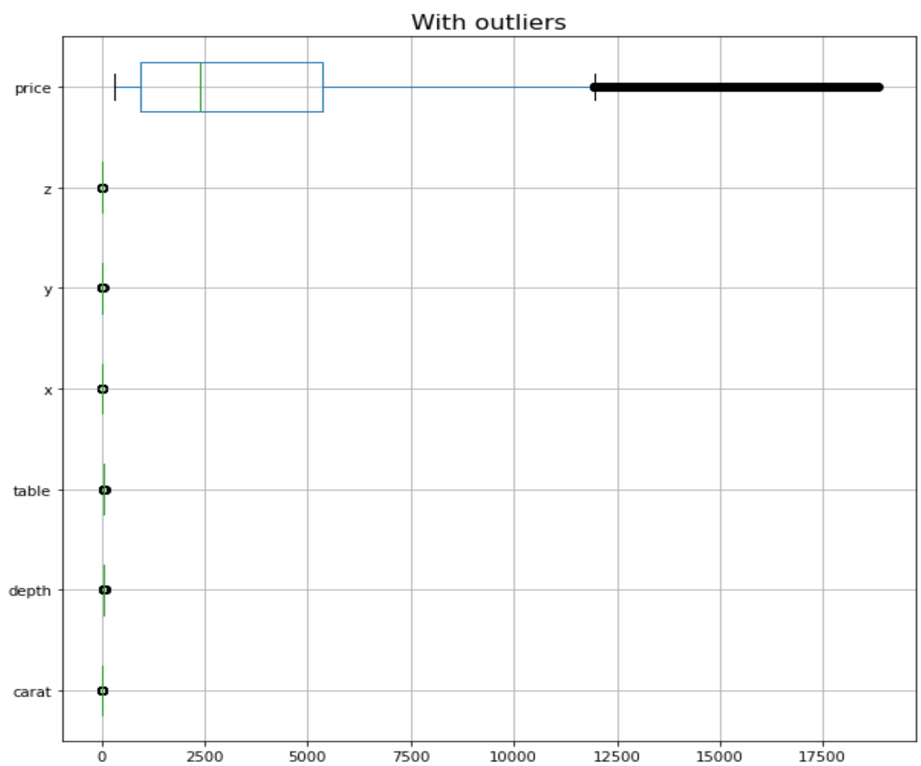

Fig 1. Boxplot problem 1

From the above boxplot we can infer that outliers are present in the dataset in all the continuous variables depicted above but maximum outliers are present in the price column.

Pairplot

Pairplot shows the relationship between the variables in the form of scatterplot and the distribution of the variable in the form of histogram or density function.

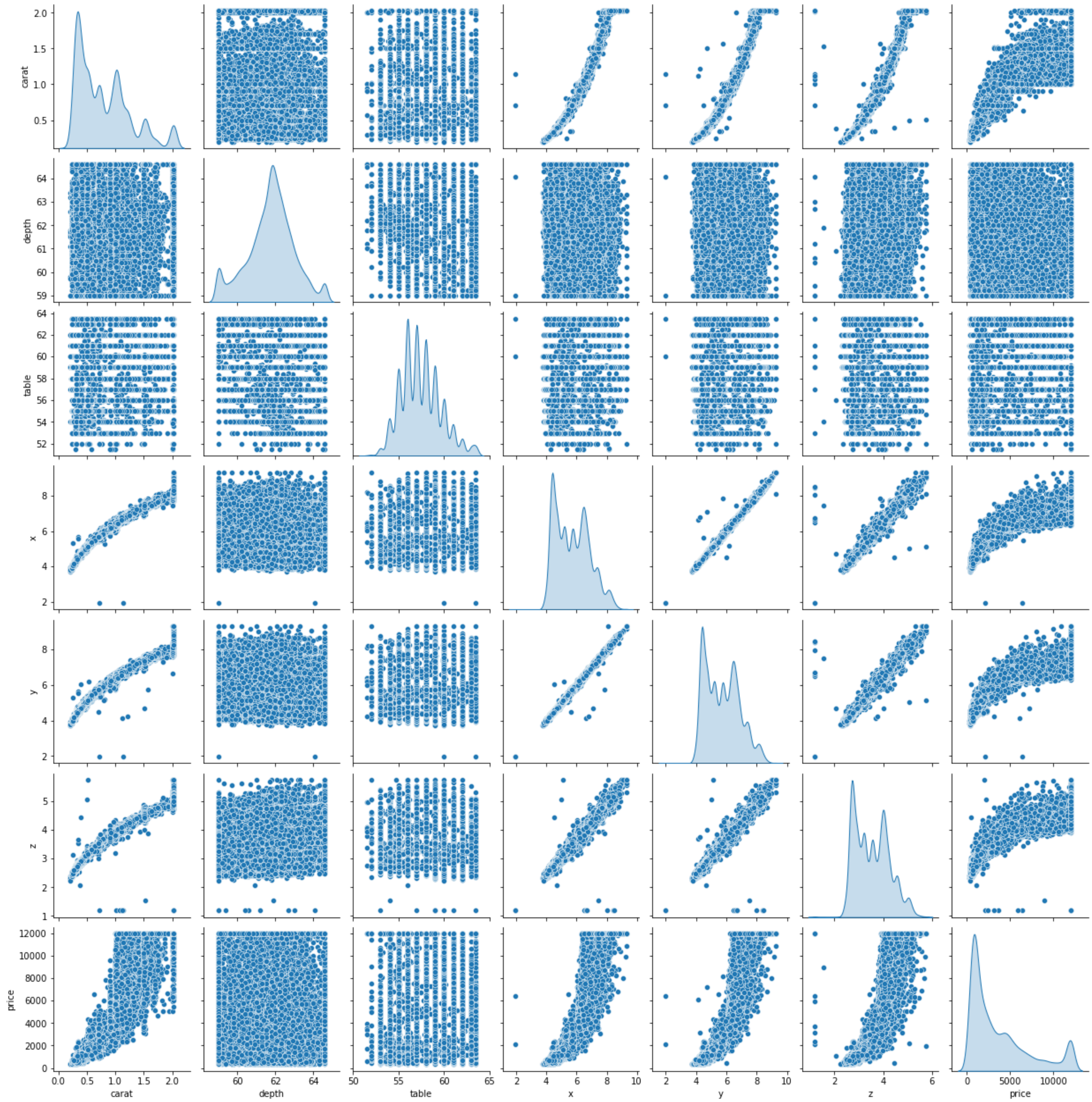

Fig 2. Pairplot problem 1

From the above pairplot we can infer that :

- There is a positive correlation between x and carat, y and carat and z and carat.
- There is very high degree of positive correlation between x and z , y and z.
- Price is also positively correlated with carat, x, y and z.

Correlation heatmap

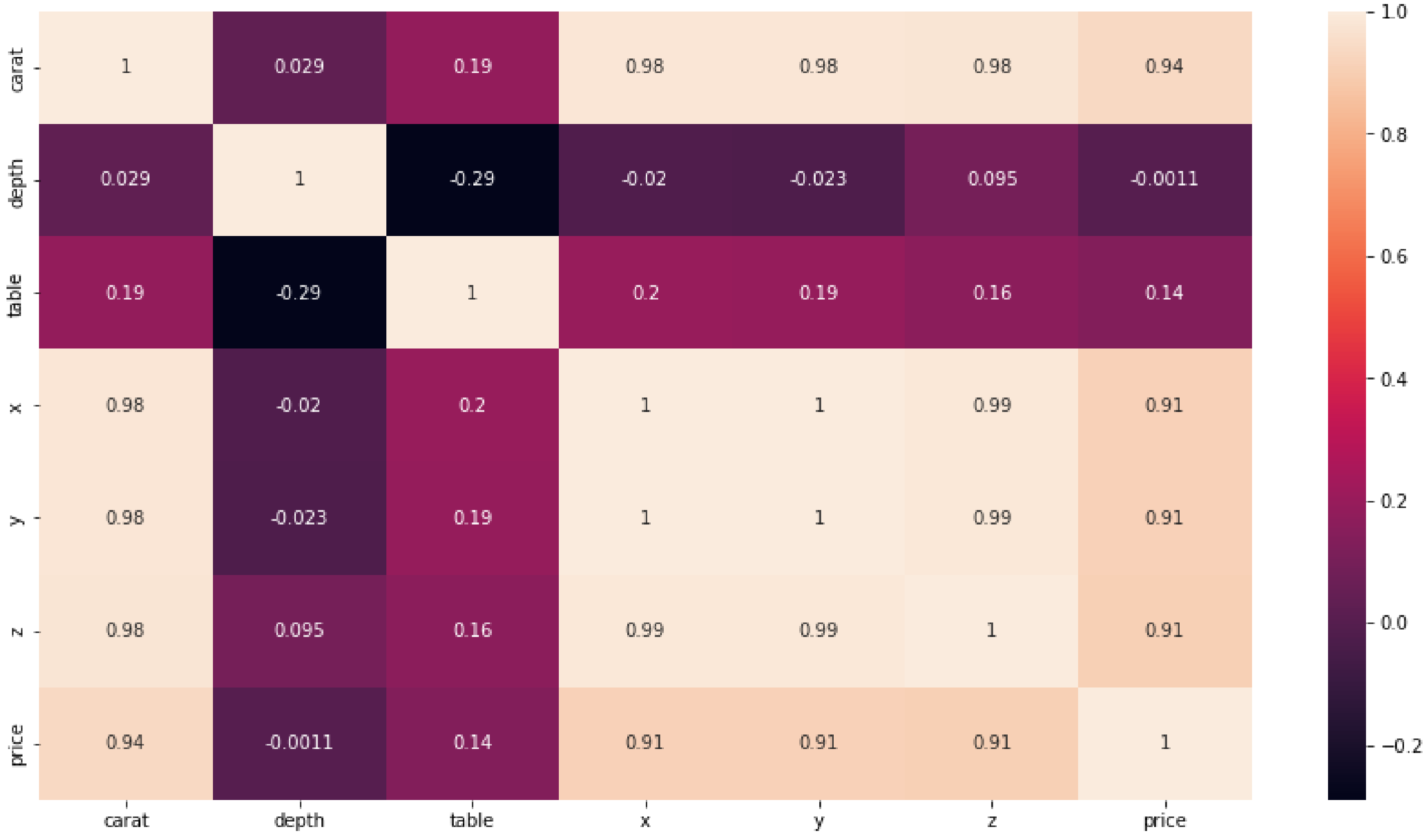

Fig 3. Correlation heatmap problem 1

From the above correlation heatmap we can see that the correlation between z and x, z and y is 0.99 while the x,y,z are also highly correlated with carat. Similarly price has correlation of 0.94 with carat and 0.91 with x,y and z.

1.2 Impute null values if present, also check for the values which are equal to zero. Do they have any meaning or do we need to change them or drop them? Check for the possibility of combining the sub levels of ordinal variables and take actions accordingly. Explain why you are combining these sub levels with appropriate reasoning.

- The null values are present in “depth” column so we will be imputing it with median.

As depth is a numerical column so either mean or median can be chosen for imputation.

After imputation with median the count of null values is as follows :

carat 0
cut 0

color	0
clarity	0
depth	0
table	0
x	0
y	0
z	0
price	0

Hence, no null value is present in the dataset.
Also, there is no value in the dataset having value as 0.

- The various sub-levels in the categorical variables are :
  CUT : 5 sub-levels in the increasing order i.e. Fair, Good, Very Good, Premium and Ideal

  COLOR : 7 sub-levels with D being the worst and J being the best i.e. D, E, F, G, H, I, J

  CLARITY : 8 sub-levels from worst to best in terms of average price i.e IF, VVS1, VVS2, VS1, VS2, SI1, SI2, I1

  - ❖ We are combining the sub-levels i.e Good and Very Good as Good in "cut" attribute.
  - ❖ We are combining VVS1 and VVS2 as VVS, VS1 and VS2 as VS and SI1, SI2 as SI in "clarity" attribute.
  - ❖ These levels are combined to reduce the dimensions of the dataset.
- After combining the various sub-levels the value counts of the above categorical variables are as follows :

CUT		CLARITY	
Ideal	10805	SI	11129
Good	8462	VS	10180
Premium	6886	VVS	4369
Fair	780	IF	891
		I1	364

Table 4. Combined sub-levels

1.3 Encode the data (having string values) for Modelling. Split the data into train and test (70:30). Apply Linear regression using scikit learn. Perform checks for significant variables using appropriate method from statsmodel. Create multiple models and check the performance of Predictions on Train and Test sets using Rsquare, RMSE & Adj Rsquare. Compare these models and select the best one with appropriate reasoning.

For encoding the data having string values we are doing ordinal encoding as data is given in a proper order.

- Values are assigned from 1 to 4 in "cut" attribute where 1 being fair and 4 being ideal.
- Values are assigned from 1 to 7 in "color" attribute where 1 is given to D (being worst) and 7 is given to J (being best).
- Values are assigned from 1 to 5 in "clarity" attribute where 1 being worst i.e. IF and 5 being best i.e. I1

Linear Regression Model

For building linear regression model the data is splitted in 70:30 and random_state is taken as 1.
Here "price" is our target variable.

Model 1 : Without outliers and combined sub-levels

The intercept for our model is 8294.302886318774
The coefficient for carat is 9515.452855470348
The coefficient for cut is 185.1714003413963
The coefficient for color is -112.1478114638456
The coefficient for clarity is -112.14781146385388
The coefficient for depth is -76.01711350781625
The coefficient for table is -37.62923797937327
The coefficient for x is -2866.652959975616
The coefficient for y is 2191.4532974919716
The coefficient for z is -320.5708934410865

Model 1 performance :

R^2 train = 0.895
R^2 test = 0.898
Adjusted r-squared : 0.895
RMSE train = 1122.517
RMSE test = 1106.187

Stats model summary

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.895
Model:                            OLS   Adj. R-squared:                  0.895
Method:                 Least Squares   F-statistic:                 2.010e+04
Date:                Sun, 03 Jul 2022   Prob (F-statistic):               0.00
Time:                        10:08:12   Log-Likelihood:            -1.5916e+05
No. Observations:               18853   AIC:                         3.183e+05
Df Residuals:                   18844   BIC:                         3.184e+05
Df Model:                           8
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   8294.3029    851.670      9.739      0.000    6624.952    9963.653
carat       9515.4529     99.912     95.239      0.000    9319.617    9711.289
cut          185.1714     10.911     16.971      0.000     163.784     206.559
color       -112.1478      2.516    -44.576      0.000    -117.079    -107.217
clarity     -112.1478      2.516    -44.576      0.000    -117.079    -107.217
depth        -76.0171     11.021     -6.897      0.000     -97.619     -54.415
table        -37.6292      4.818     -7.810      0.000     -47.073     -28.185
x          -2866.6530    146.992    -19.502      0.000   -3154.770   -2578.536
y           2191.4533    145.705     15.040      0.000    1905.859    2477.048
z           -320.5709    118.339     -2.709      0.007    -552.526     -88.616
==============================================================================
Omnibus:                     4919.617   Durbin-Watson:                   2.005
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            25089.758
Skew:                           1.169   Prob(JB):                         0.00
Kurtosis:                       8.146   Cond. No.                     2.79e+17
```

Table 5. Stats model 1 summary

The overall p value is less than alpha (0.05) so rejecting null hypothesis and accepting alternate hypothesis that atleast 1 regression coefficient is not 0. Here all regression coefficients are not 0.

Model 2 : With outliers and combined sub-levels

The intercept for our model is 16687.925289551575
The coefficient for carat is 11581.016683733771
The coefficient for cut is 181.21235755877842

The coefficient for color is -135.66237812264438
The coefficient for clarity is -135.6623781226433
The coefficient for depth is -153.67557438776942
The coefficient for table is -63.74257026343272
The coefficient for x is -1484.4337011028053
The coefficient for y is 19.24013001963791
The coefficient for z is -3.532364468217622

Model 2 performance :
R^2 train = 0.872
R^2 test = 0.875
Adjusted r-squared : 0.873
RMSE train = 1429.97
RMSE test = 1427.15

Stats model summary

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.873
Model:                            OLS   Adj. R-squared:                  0.873
Method:                 Least Squares   F-statistic:                 1.616e+04
Date:                Sun, 03 Jul 2022   Prob (F-statistic):               0.00
Time:                        22:34:46   Log-Likelihood:            -1.6373e+05
No. Observations:               18853   AIC:                         3.275e+05
Df Residuals:                   18844   BIC:                         3.275e+05
Df Model:                           8
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   1.669e+04    819.832     20.355      0.000    1.51e+04    1.83e+04
carat       1.158e+04    107.535    107.695      0.000    1.14e+04    1.18e+04
cut          181.2124     13.806     13.126      0.000     154.152     208.273
color       -135.6624      3.206    -42.317      0.000    -141.946    -129.379
clarity     -135.6624      3.206    -42.317      0.000    -141.946    -129.379
depth       -153.6756      9.064    -16.954      0.000    -171.443    -135.908
table        -63.7426      5.856    -10.886      0.000     -75.220     -52.265
x          -1484.4337     58.108    -25.546      0.000   -1598.330   -1370.537
y             19.2401     28.033      0.686      0.493     -35.708      74.188
z             -3.5324     46.695     -0.076      0.940     -95.059      87.994
==============================================================================
Omnibus:                     4628.916   Durbin-Watson:                   1.997
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           128134.654
Skew:                           0.573   Prob(JB):                         0.00
Kurtosis:                      15.720   Cond. No.                     3.51e+17
```

Table 6. stats model 2 summary

<u>Model 3 : Without outliers and no combined sub-levels</u>
The intercept for our model is 2695.3998518633825
The coefficient for carat is 8829.985689983425
The coefficient for cut is 110.86409383215617
The coefficient for color is -278.00475388207354
The coefficient for clarity is -440.56098957937377
The coefficient for depth is -7.366929069916139
The coefficient for table is -12.335846538458245
The coefficient for x is -1412.5450184859153
The coefficient for y is 1247.8206514842923

The coefficient for z is -313.13928882800053

Model 3 performance :
R^2 train = 0.931
R^2 test = 0.931
Adjusted r-squared : 0.94
RMSE train = 909.245
RMSE test = 911.142

Stats model summary

OLS Regression Results

Dep. Variable:	price	R-squared:	0.940
Model:	OLS	Adj. R-squared:	0.940
Method:	Least Squares	F-statistic:	1.287e+04
Date:	Sun, 03 Jul 2022	Prob (F-statistic):	0.00
Time:	22:52:47	Log-Likelihood:	-1.5386e+05
No. Observations:	18853	AIC:	3.078e+05
Df Residuals:	18829	BIC:	3.080e+05
Df Model:	23		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	2152.7206	657.170	3.276	0.001	864.608	3440.833
cut[T.2]	481.8102	44.224	10.895	0.000	395.128	568.492
cut[T.3]	606.9286	42.338	14.335	0.000	523.942	689.915
cut[T.4]	674.8261	41.377	16.309	0.000	593.723	755.929
cut[T.5]	714.6999	43.069	16.594	0.000	630.280	799.120
color[T.2]	-181.9043	22.750	-7.996	0.000	-226.496	-137.312
color[T.3]	-256.8104	23.219	-11.060	0.000	-302.323	-211.298
color[T.4]	-429.3834	22.528	-19.060	0.000	-473.540	-385.227
color[T.5]	-855.9903	24.109	-35.505	0.000	-903.246	-808.734
color[T.6]	-1323.9249	26.806	-49.389	0.000	-1376.467	-1271.383
color[T.7]	-1928.0464	33.046	-58.345	0.000	-1992.819	-1863.274
clarity[T.2]	-231.7148	41.598	-5.570	0.000	-313.251	-150.179
clarity[T.3]	-246.2406	39.913	-6.170	0.000	-324.473	-168.008
clarity[T.4]	-661.4432	38.199	-17.315	0.000	-736.318	-586.569
clarity[T.5]	-964.0870	37.346	-25.815	0.000	-1037.288	-890.886
clarity[T.6]	-1484.1033	37.586	-39.486	0.000	-1557.774	-1410.432
clarity[T.7]	-2319.5585	39.080	-59.354	0.000	-2396.159	-2242.958
clarity[T.8]	-4004.0199	66.185	-60.497	0.000	-4133.749	-3874.291
carat	9126.8384	76.199	119.777	0.000	8977.482	9276.195
depth	-14.9411	8.512	-1.755	0.079	-31.625	1.742
table	-18.5773	3.843	-4.834	0.000	-26.110	-11.045
x	-1190.1742	120.504	-9.877	0.000	-1426.372	-953.976
y	837.6187	120.630	6.944	0.000	601.173	1074.064
z	-164.1665	89.799	-1.828	0.068	-340.181	11.848

Table 7. stats model 3 summary

Conclusion :

- Model 3 performance is better compared to other models.
- The adjusted r-squared for model 3 is 0.94 i.e. 94% of the variability in price is explained by all the other attributes.
- The overall p value is less than alpha (0.05) so rejecting null hypothesis and accepting alternate hypothesis that atleast 1 regression coefficient is not 0. Here all regression coefficients are not 0.

1.4 Inference: Basis on these predictions, what are the business insights and recommendations.

- Carat has a larger value of coefficient so it will have more effect on price of cubic zirconia.
- R-squared value in model 3 is 0.94 which shows that 84% variance in price is explained by all the other attributes.

- There are some negative coefficient values for instance depth has its corresponding coefficient as -14.9. This implies, that the price decreases by 14.9 units keeping all other predictors same.
- As the cut quality increases the price of the diamond also increases.
- The width (y) of the cubic zirconia also have positive impact on price of the diamond.

PROBLEM 2 : LOGISTIC REGRESSION AND LDA.

You are hired by a tour and travel agency which deals in selling holiday packages. You are provided details of 872 employees of a company. Among these employees, some opted for the package and some didn't. You have to help the company in predicting whether an employee will opt for the package or not on the basis of the information given in the data set. Also, find out the important factors on the basis of which the company will focus on particular employees to sell their packages.

DATA ATTRIBUTES

Variable Name	Description
Holiday_Package	Opted for Holiday Package yes/no?
Salary	Employee salary
age	Age in years
edu	Years of formal education
no_young_children	The number of young children (younger than 7 years)
no_older_children	Number of older children
foreign	foreigner Yes/No

SAMPLE OF THE DATASET

	Unnamed: 0	Holliday_Package	Salary	age	educ	no_young_children	no_older_children	foreign
0	1	no	48412	30	8	1	1	no
1	2	yes	37207	45	8	0	1	no
2	3	no	58022	46	9	0	0	no
3	4	no	66503	31	11	2	0	no
4	5	no	66734	44	12	0	2	no

Table 8. Original dataset sample problem 2

The dataset have 8 variables but Unnamed column is not useful for building the model so we will drop it.

2.1 Data Ingestion: Read the dataset. Do the descriptive statistics and do null value condition check, write an inference on it. Perform Univariate and Bivariate Analysis. Do exploratory data analysis.

Check for the shape of the dataset.

After dropping the Unnamed column the shape of the dataset is (872,7) i.e. there are 872 rows and 7 columns.

Check for the data type.

Holliday_Package	object
Salary	int64
age	int64

educ int64
no_young_children int64
no_older_children int64
foreign object
There are 2 object data type and others are integer data type.

Check for the missing values in the dataset.

Holliday_Package 872 non-null
Salary 872 non-null
age 872 non-null
educ 872 non-null
no_young_children 872 non-null
no_older_children 872 non-null
foreign 872 non-null
There are no null values in the given dataset.

Check for duplicates

There are no duplicates present in the dataset.

Summary stats

Descriptive statistics help describe and understand the features of a specific dataset by giving short summaries about the sample and measures of the data. The most recognized types of descriptive statistics are measures of centre : the mean, median and mode.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Holliday_Package	872	2	no	471	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Salary	872.0	NaN	NaN	NaN	47729.172018	23418.668531	1322.0	35324.0	41903.5	53469.5	236961.0
age	872.0	NaN	NaN	NaN	39.955275	10.551675	20.0	32.0	39.0	48.0	62.0
educ	872.0	NaN	NaN	NaN	9.307339	3.036259	1.0	8.0	9.0	12.0	21.0
no_young_children	872.0	NaN	NaN	NaN	0.311927	0.61287	0.0	0.0	0.0	0.0	3.0
no_older_children	872.0	NaN	NaN	NaN	0.982798	1.086786	0.0	0.0	1.0	2.0	6.0
foreign	872	2	no	656	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Table 9. Summary (problem 2)

From the above descriptive summary we can infer that the average salary of the people is around 47729 while the maximum is 236961. People not taking holiday package are 471 and 656 foreigners are also not availing the holiday package.

Boxplot

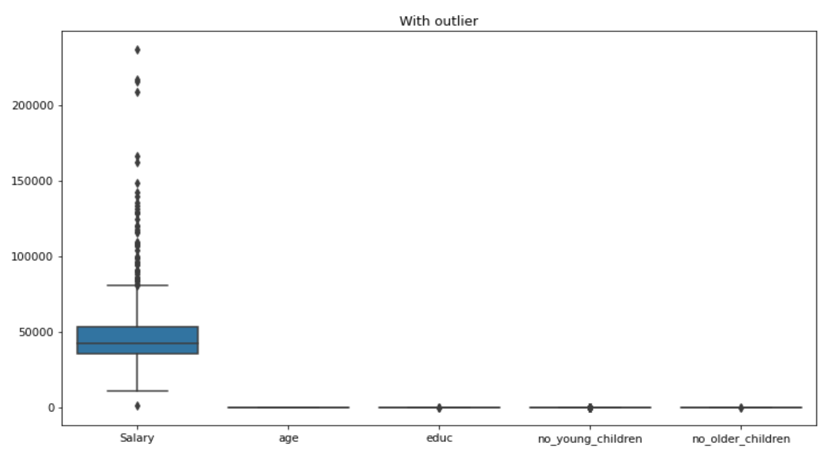

Fig 4. Boxplot problem 2

From the above boxplot we can infer that there are many outliers present in Salary column while educ, no_young_children and no_older_children also has few outliers.

Pairplot

Pairplot shows the relationship between the variables in the form of scatterplot and the distribution of the variable in the form of histogram or density function.

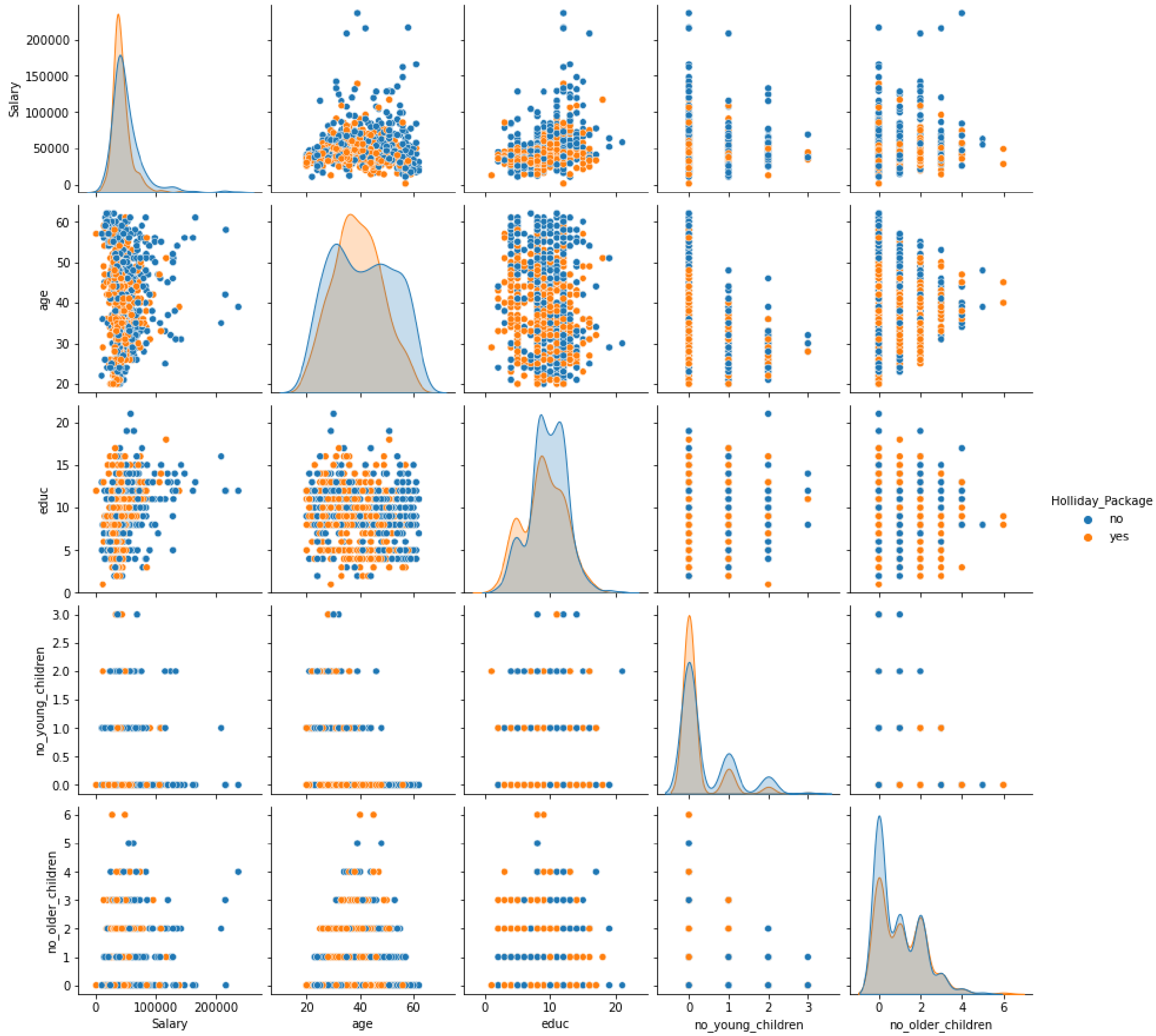

Fig 5. Pairplot problem 2

The above pairplot is created by taking “Holliday package” as hue. We can infer that there is very less or no correlation between the variables.

Correlation heatmap

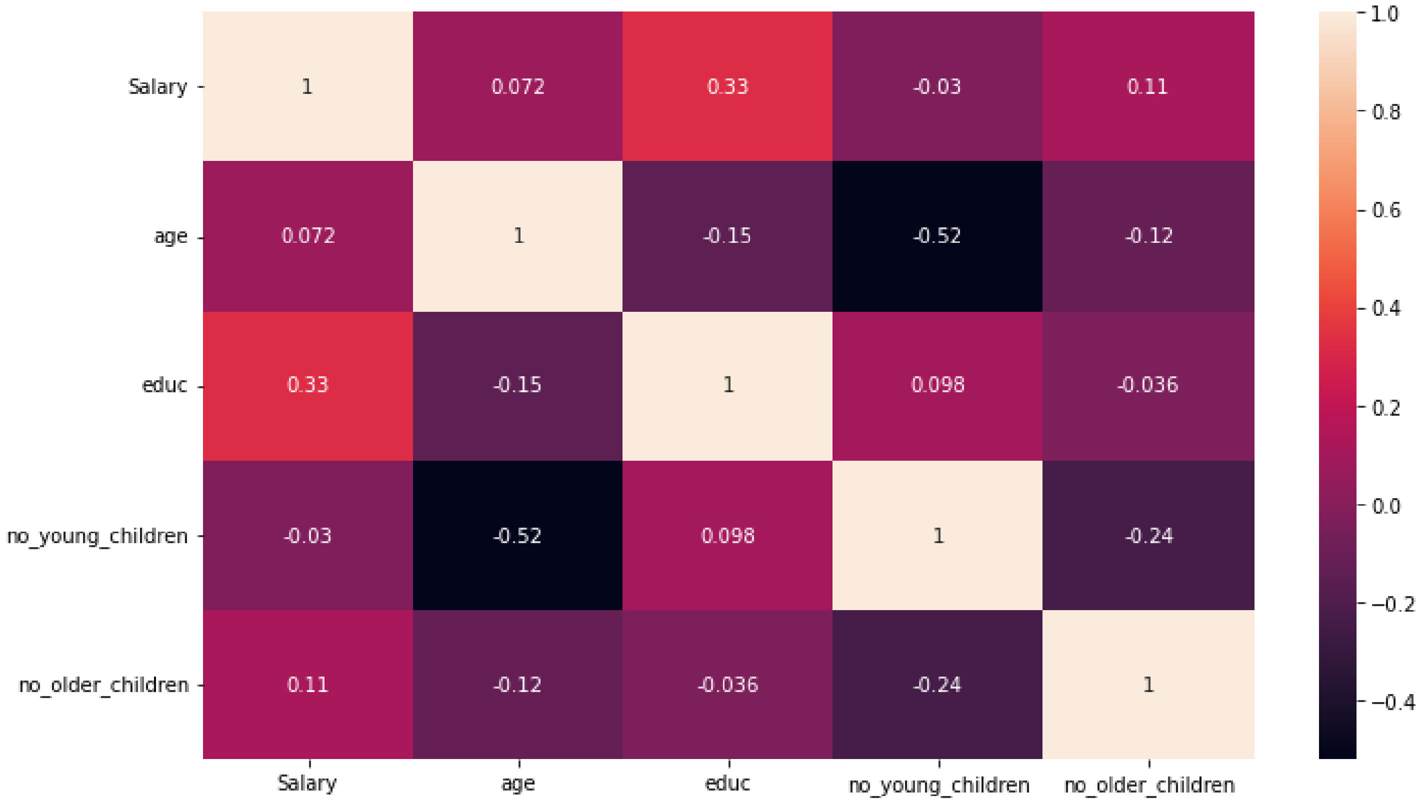

Fig 6. Correlation heatmap problem 2

From the above heatmap we can infer that there is very less or no correlation between the variables

Count plot

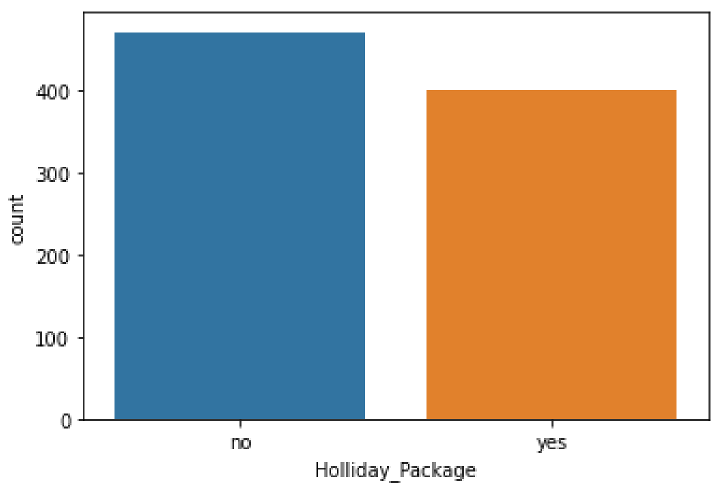

Fig 7. Countplot

The above count plot depicts the count of Holliday package i.e. people who opted for the package (yes) and who didn't opt for the package (no).
Approximately 470 didn't opt for the package while around 400 opted for it.

2.2 Do not scale the data. Encode the data (having string values) for Modelling. Data Split: Split the data into train and test (70:30). Apply Logistic Regression and LDA (linear discriminant analysis).

We are applying label encoding to the variables having string values.

- People who opted for holliday package (yes) are coded as 1 and the ones who didn't opted for it (no) are coded as 0.
- People who are foreigners (yes) are coded as 1 and the ones who are not foreigners (no) are coded as 0.

For building both the models i.e. Logistic Regression and LDA the data is splitted into 70:30.
The value counts of the training set is :

```
0    0.534
1    0.465
```

The value counts of the testing set is :

```
0    0.553435
1    0.446565
```

Logistic Regression model :

- Here Holliday_Package is the target variable.
- The model is build without treating the outliers.
- The attributes taken for building the model are max_iter as 10000 and n_jobs = 2, rest all are default values.

LDA :

- Holliday_Package is the target variable.
- The model is build with outliers.
- The attributes taken are default values.

2.3 Performance Metrics: Check the performance of Predictions on Train and Test sets using Accuracy, Confusion Matrix, Plot ROC curve and get ROC_AUC score for each model Final Model: Compare both the models and write inference which model is best/optimized.

Logistic Regression model performance :

Training set :

- Accuracy score : 0.519
- Roc_auc score : 0.567

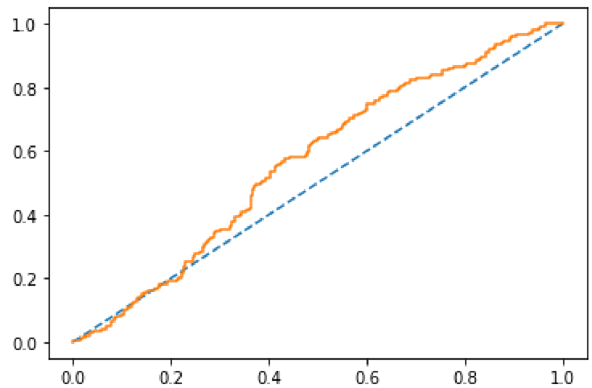

Fig 8. Roc curve training (logistic regression)

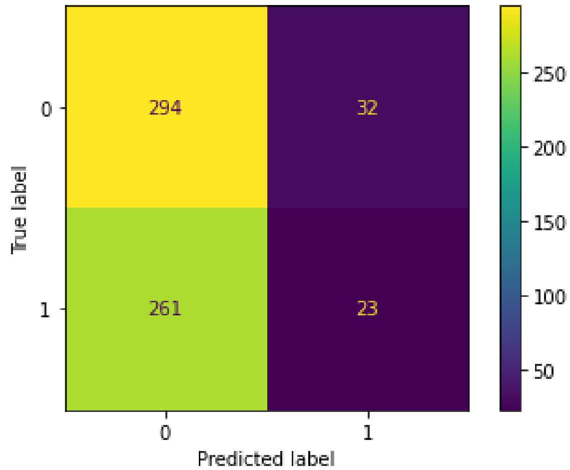

Fig 9. Confusion matrix training (logistic reg)

From the above confusion matrix we can see that false positives are 261 and false negatives are 32

	precision	recall	f1-score	support
0	0.53	0.90	0.67	326
1	0.42	0.08	0.14	284
accuracy			0.52	610
macro avg	0.47	0.49	0.40	610
weighted avg	0.48	0.52	0.42	610

Table 9. Classification report training (logistic reg)

The recall score is 8% for the positive class in hand which is not good estimator while the precision is 42% and f1 score is 14% for the positive class in hand.

<u>Testing set :</u>

- Accuracy score : `0.530`
- Roc_auc score : `0.627`

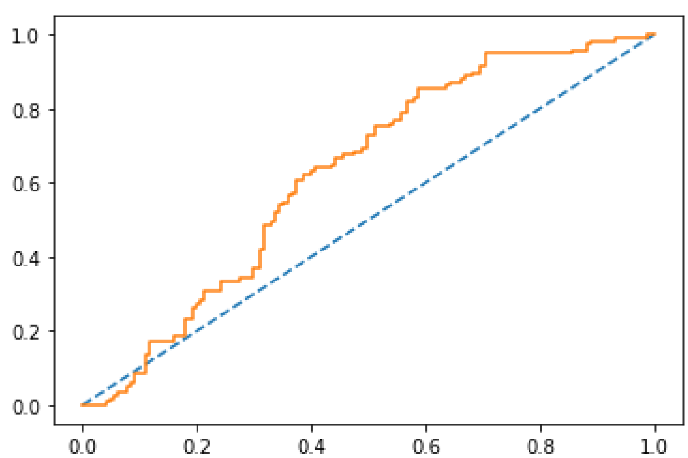

Fig 10. Roc curve testing (logistic reg)

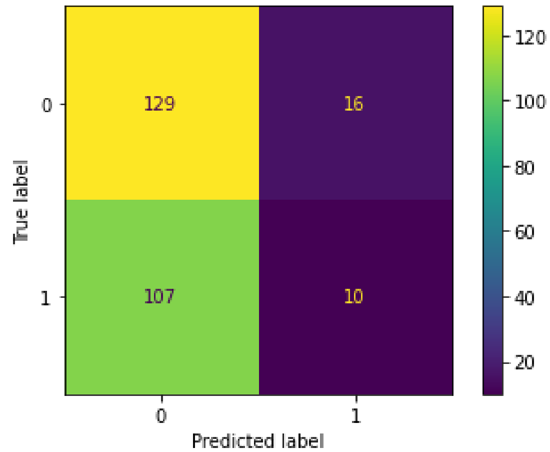

Fig 11. Confusion matrix testing (logistic reg)

The value of false positives is 107 while false negatives is 16 in testing set.

	precision	recall	f1-score	support
0	0.55	0.89	0.68	145
1	0.38	0.09	0.14	117
accuracy			0.53	262
macro avg	0.47	0.49	0.41	262
weighted avg	0.47	0.53	0.44	262

Table 10. Classification report testing (logistic reg)

The recall of the testing data is 0.09 for positive class in hand while precision is 0.38 and f1 score is 0.14 for positive class in hand.

LDA model performance :

Training set :

- Accuracy score : 0.672
- Roc_auc score : 0.742

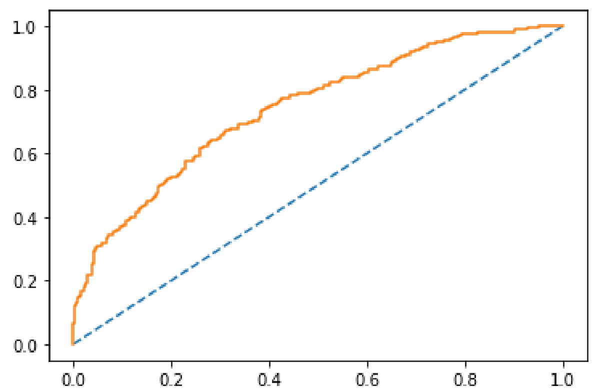

Fig 12. Roc curve training (LDA)

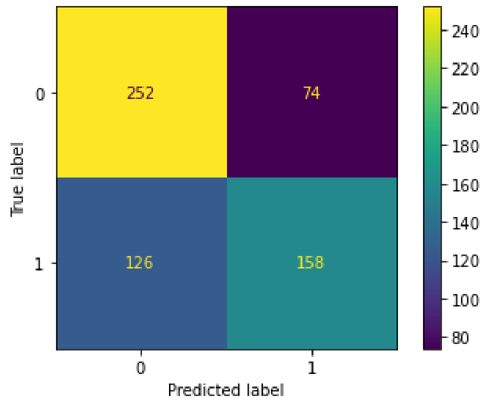

Fig 13. Confusion matrix training (LDA)

The value of false positives in training set is 126 while that of false negative is 74.

	precision	recall	f1-score	support
0	0.67	0.77	0.72	326
1	0.68	0.56	0.61	284
accuracy			0.67	610
macro avg	0.67	0.66	0.66	610
weighted avg	0.67	0.67	0.67	610

Table 11. Classification report training (LDA)

The recall of the positive class in hand in training set is 0.56 while precision is 0.68 and f1 score is 0.61.

<u>Testing set :</u>

- Accuracy score : 0.641
- Roc_auc score : 0.703

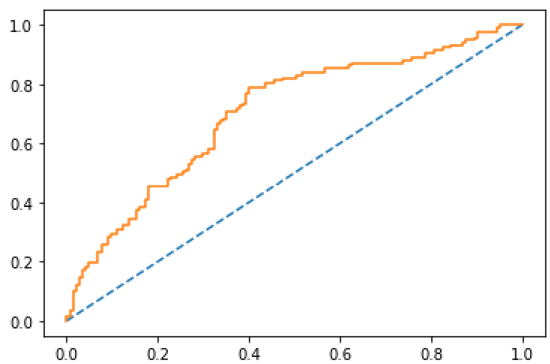

Fig 14. Roc curve testing (LDA)

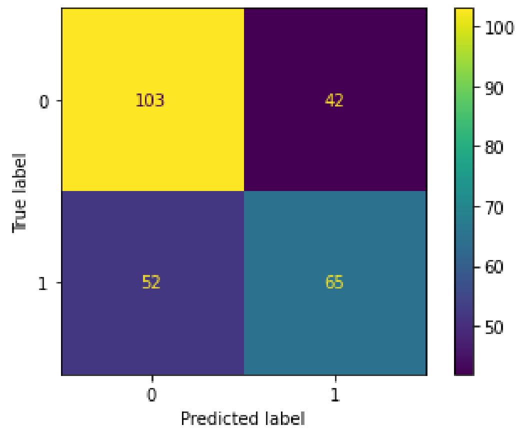

Fig 15. Confusion matrix testing (LDA)

The value of false positives in testing set is 52 while that of false negative is 42.

	precision	recall	f1-score	support
0	0.66	0.71	0.69	145
1	0.61	0.56	0.58	117
accuracy			0.64	262
macro avg	0.64	0.63	0.63	262
weighted avg	0.64	0.64	0.64	262

Table 12. Classification report testing (LDA)

The recall of the positive class in hand in testing data is 0.56 while precision is 0.61 and f1 score is 0.58.

Model comparison :

- Accuracy, AUC, Precision and recall for test data is almost inline with training data in both the models thus, there is no overfitting or underfitting.
- The accuracy and recall for LDA is much higher than that of Logistic Regression thus, LDA should be used for classification.
- The roc score for testing data in logistic regression is around 64% while in LDA it is 70%
- The above scores are not affected by the class imbalance problem as proportion is almost same in training and testing set.

2.4 Inference: Basis on these predictions, what are the insights and recommendations.

Business insights and recommendations :

- Here the metric Recall is most important so the major focus can be upon imporving the recall score.
- The correlation between the variables is very less.
- From the confusion matrix of test data of LDA model we can see that our model has misclassified 42 employees as they have taken the holliday package while they were predicted as not taking it.
- The agency can give various offers and promotional schemes to employees within a specific salary range.